

A CERTIFIED VELOCITY-RESIDUAL DETECTOR FOR OFF-AXIS ZEROS OF THE RIEMANN Ξ -FUNCTION, AND THE DE BRUIJN–NEWMAN FLOW AS A MATRIX PENCIL ON CALOGERO–MOSER SPACE

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ABSTRACT. Let $\Xi(z) = \xi(\frac{1}{2} + iz)$ denote the Riemann Ξ -function. At any simple real zero γ of Ξ , the ratio $V = \Xi''(\gamma)/\Xi'(\gamma)$ equals the two-dimensional Coulomb field of the *entire* zero multiset evaluated at γ — including any hypothetical zeros off the critical line. We turn this asymmetry into a certified diagnostic. Using the exact Riemann–von Mangoldt decomposition, an exactly computable boundary term at the truncation point, and Trudgian’s unconditional bound on $S(t)$, the residual of the measured velocity against the field of the first K zeros lies in a rigorously certified band; at $K = 1000$ the observed residuals sit four orders of magnitude below that band. A multipole sign discriminant then distinguishes off-axis from on-axis hidden configurations: the leading correction to the monopole field is a quadrupole whose *sign* records whether the hidden displacement is perpendicular or parallel to the critical line. The certified sensitivity covers the entire parameter range permitted by the current bound $\Lambda \leq 0.2$ on the De Bruijn–Newman constant. Detected configurations feed a linear matrix pencil $X + tY$ on the Calogero–Moser space whose real-spectrum threshold is the local De Bruijn–Newman margin: we prove that the backward heat flow on the roots of a polynomial is exactly the straight line $t \mapsto X + tY$ — a De Bruijn–Newman reading of Wilson’s tau-function formula for rational solutions of the KP hierarchy — so that $\Lambda(p)$ is the reality threshold of a linear eigenvalue problem. The method uses only the Hadamard product structure of Ξ and therefore ports verbatim to self-dual Selberg-class L -functions; we demonstrate it on the Dirichlet beta function. Everything is unconditional: the Riemann hypothesis is assumed nowhere.

1. INTRODUCTION

1.1. **Setting.** Write $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ for the completed zeta function and

$$\Xi(z) = \xi\left(\frac{1}{2} + iz\right),$$

so that Ξ is entire, even, real on the real axis, of order one, and its zero multiset $\{\pm\rho_k\}$ consists of the nontrivial zeros of ζ rotated onto (a neighbourhood of) the real axis. The Riemann hypothesis (RH) is the statement that every ρ_k is real. Since $\sum_k |\rho_k|^{-2} < \infty$, the Hadamard product is genus zero in z^2 :

$$(1) \quad \Xi(z) = \Xi(0) \prod_{k \geq 1} \left(1 - \frac{z^2}{\rho_k^2}\right).$$

Everything in this paper uses only (1) and elementary complex analysis, which is why it transfers without change to any self-dual element of the Selberg class (Section 6).

Two classical circles of ideas frame our results. The first is the *heat-flow deformation* of de Bruijn [4] and Newman [17]: setting $H_t = e^{-t\partial_z^2} H_0$ with $H_0 = \Xi$ (equivalently, inserting the

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factor e^{tu^2} under Riemann’s integral representation), there is a constant Λ , the De Bruijn–Newman constant, such that H_t has only real zeros if and only if $t \geq \Lambda$. RH is the statement $\Lambda \leq 0$; Rodgers and Tao proved $\Lambda \geq 0$ [23], and the Polymath15 project [22], combined with the verification of RH to height $3 \cdot 10^{12}$ by Platt and Trudgian [20], gives $\Lambda \leq 0.2$. Under the flow, the real zeros move by the first-order dynamics

$$(2) \quad \dot{z}_j = \sum_{k \neq j} \frac{2}{z_j - z_k},$$

that is, as a one-dimensional gas of logarithmically repelling charges [23, Theorem 11]; the same dynamics is classical in the integrable-systems literature as the pole motion of rational solutions of the KP hierarchy [1, 14, 10].

The second circle is *numerical verification*. Since Turing [25], completeness of a list of critical-line zeros below a height T is certified by counting: an argument-principle count of all zeros is compared with the number found on the line (see [3] for a survey and [20, 19] for the state of the art). Turing’s method reports a discrepancy only as “a zero is missing below T ”: it produces no location, no distance from the line, and no quantitative margin.

1.2. What this paper does. This is a methods paper. Its main object is a diagnostic that upgrades a counting discrepancy into a diagnosis, and its main theoretical ingredient is an exact linear-algebraic model of the De Bruijn–Newman flow. The results are unconditional throughout; “certified” means rigorous modulo floating-point arithmetic (see Section 7).

The detector (Sections 2–4). At any simple real zero γ_j of Ξ , the quantity

$$V_j = \frac{\Xi''(\gamma_j)}{\Xi'(\gamma_j)}$$

is computable to arbitrary precision from ζ alone, and it equals the Coulomb field $\sum_{\rho \neq \gamma_j} 2/(\gamma_j - \rho)$ of the *full* zero multiset — visible zeros, hidden off-axis zeros, and real zeros absent from the list alike (Lemma 2.1). Subtracting the field of the first K listed zeros and a certified tail leaves a residual R_j that must lie in an explicit band $\pm \varepsilon_j$ if the list is complete. The band is made rigorous by an exact bookkeeping of the Riemann–von Mangoldt formula: the boundary term $S(U)$ at the truncation point is *exactly* computable there, and the remaining oscillatory integral is majorized by Trudgian’s unconditional bound $|S(u)| \leq 0.111 \log u + 0.275 \log \log u + 2.450$ [24]. At $K = 1000$ the observed residuals are $\sim 10^{-7}$ against certified bands $\sim 10^{-3}$ (Table 1).

A hidden configuration announces itself in R_j through its multipole expansion, and the leading distinguishing term carries a sign (Lemma 2.2): an off-axis quadruple $\pm(\tau \pm i\sigma)$ contributes $4/d - 4\sigma^2/d^3 + O(d^{-5})$ at distance d from its centre, an on-axis pair $\pm(\tau \pm a)$ contributes $4/d + 4a^2/d^3 + O(d^{-5})$. The monopole certifies *existence* and locates the centre; the quadrupole’s sign reads off whether the hidden displacement is perpendicular or parallel to the axis. In certified terms, at $K = 1000$ any hidden pair within distance $\approx 1.8 \cdot 10^3$ of a measured zero is visible, and off-/on-axis discrimination is certified down to $\sigma \approx 0.024$ at unit distance — while the bound $\Lambda \leq 0.2$ confines any hypothetical off-axis zero to $\sigma \lesssim 0.63$ (up to interaction corrections computed in Section 5.2), entirely inside the certified window.

The pencil (Section 5). For a monic polynomial p of degree n with distinct roots $z_1, \dots, z_n$, form the Calogero–Moser pair

$$X = \text{diag}(z_1, \dots, z_n), \quad Y_{jk} = \frac{2}{z_j - z_k} \ (j \neq k), \quad Y_{jj} = \sum_{k \neq j} \frac{2}{z_j - z_k}.$$

Then (Theorem 5.1) for all $t \in \mathbb{C}$,

$$(e^{-t\partial_z^2}p)(z) = \det(zI - X - tY).$$

The identity itself is not new: it is George Wilson’s tau-function formula for rational solutions of the KP hierarchy [26], stated in exactly this determinantal form in [9, §6.3], with the heat flow appearing as the second KP time; its expectation over a Gaussian ensemble is the finite free probability of Marcus, Spielman and Srivastava [12] (see also [8]). What we believe is new is the De Bruijn–Newman reading and its two corollaries: the De Bruijn–Newman constant of a real polynomial is the *real-spectrum threshold of a linear matrix pencil*,

$$\Lambda(p) = \inf\{t : \text{spec}(X + tY) \subset \mathbb{R}\},$$

and the De Bruijn–Newman flow — the object at the centre of Newman’s conjecture, the Rodgers–Tao theorem, and the Polymath15 computations — is a *straight line* on Calogero–Moser space. This gives the detector its fourth output: a detected configuration, together with its local environment of visible zeros, feeds the pencil, and the pencil’s reality threshold is the interaction-corrected local De Bruijn–Newman margin — the off-axis counterpart of the Lehmer-pair machinery of Csordas, Smith and Varga [5].

1.3. Why a physicist might care. The space on which the pencil lives is not an exotic auxiliary object. The completed phase space of the rational Calogero–Moser system, $\mathcal{C}_n = \{(X, Z) : \text{rank}([X, Z] - I) = 1\} // \text{GL}_n(\mathbb{C})$ [26, 10], is the same variety that arises in gauge theory as the moduli space of charge- n $U(1)$ instantons on noncommutative $\mathbb{R}^4$: the noncommutative deformation of the ADHM equations [2] that resolves the small-instanton singularity of the Hilbert scheme $\text{Hilb}^n(\mathbb{C}^2)$ [16, 15]; see [6] for a review of the ADHM construction as a hyper-Kähler quotient. The statement proved in Section 5 can therefore be read as: *the De Bruijn–Newman heat flow on root configurations is free (straight-line) motion on the noncommutative-instanton moduli space, in the direction of the second KP flow*. We emphasise that this is a dictionary of structures, not a proof strategy; we make no claim that gauge-theoretic dynamics constrains the zeros of ζ . Its practical content in this paper is the margin meter of Section 5.2, plus the observation that a body of exact technology (projection methods, tau functions, adelic Grassmannians) applies verbatim to the flow that analytic number theory actually uses.

1.4. What this paper does not do. The detector certifies and diagnoses; it does not prove. If RH is true, it will only ever return null results, and its value is as verification infrastructure — particularly for L -functions, where verified heights are far lower than for ζ — together with the geometric dictionary it operationalises: zeros as a two-dimensional Coulomb gas whose on-axis field is exactly measurable, hidden charge as a solvable inverse-source problem, and local margins as spectra of linear pencils.

2. THE VELOCITY IDENTITY AND THE MULTIPOLE DISCRIMINANT

Throughout, $\{\rho\}$ denotes the full zero multiset of Ξ (both signs), and $\gamma_1 < \gamma_2 < \dots$ the ordinates of the zeros on the real axis (i.e. the critical line), which after the Platt–Trudgian verification [20] is provably the complete list up to height $3 \cdot 10^{12}$.

Lemma 2.1 (velocity identity). *Let γ_j be a simple real zero of Ξ . Then*

$$(3) \quad V_j := \frac{\Xi''(\gamma_j)}{\Xi'(\gamma_j)} = \sum_{\rho \neq \gamma_j} \frac{2}{\gamma_j - \rho},$$

the sum running over the full zero multiset and converging absolutely after the symmetric pairing $\rho \leftrightarrow -\rho$.

Proof. By (1), $\Xi'/\Xi(z) = \sum_k [\frac{1}{z-\rho_k} + \frac{1}{z+\rho_k}]$, the paired sum converging absolutely and locally uniformly away from zeros since $\sum |\rho_k|^{-2} < \infty$. Near the simple zero γ_j ,

$$\frac{\Xi'}{\Xi}(z) = \frac{1}{z - \gamma_j} + \sum_{\rho \neq \gamma_j} \frac{1}{\gamma_j - \rho} + O(z - \gamma_j),$$

while the Taylor expansion of Ξ'/Ξ at γ_j has constant term $\Xi''(\gamma_j)/(2\Xi'(\gamma_j))$. Comparing constant terms and multiplying by 2 gives (3). $\square$

Three readings of V_j will be used interchangeably. Analytically, it is two derivative evaluations of the completed function. Dynamically, it is the instantaneous velocity of γ_j under the backward-heat (De Bruijn–Newman) flow, by (2). Electrostatically, it is the planar Coulomb field of the full zero gas evaluated at γ_j . The left-hand side of (3) is computable to arbitrary precision from ζ alone; the right-hand side sees every zero, wherever it is. That asymmetry is the detector.

Lemma 2.2 (multipole sign discriminant). *Let a configuration of hidden zeros with centre $\tau > 0$ be added to the multiset, symmetrically in $\rho \leftrightarrow -\rho$, and let $x > 0$ be real with $d = x - \tau$, $|d|$ large compared with the configuration size. The contribution of the configuration to the field $\sum 2/(x - \rho)$ is:*

$$\begin{aligned} \text{off-axis quadruple } \pm(\tau \pm i\sigma) : \quad & \frac{4d}{d^2 + \sigma^2} = \frac{4}{d} - \frac{4\sigma^2}{d^3} + O(d^{-5}), \\ \text{on-axis pair } \pm(\tau \pm a) : \quad & \frac{2}{d-a} + \frac{2}{d+a} = \frac{4}{d} + \frac{4a^2}{d^3} + O(d^{-5}), \end{aligned}$$

in each case up to the mirror terms in $x + \tau$, which are smooth and common to both. The monopole $4/d$ is common; the first distinguishing term is the quadrupole, and its sign records whether the hidden displacement is perpendicular or parallel to the axis.

Proof. Two lines of Taylor expansion: $\frac{2}{x-\tau-i\sigma} + \frac{2}{x-\tau+i\sigma} = \frac{4d}{d^2+\sigma^2}$ and $\frac{2}{d-a} + \frac{2}{d+a} = \frac{4d}{d^2-a^2}$; expand in $1/d$. $\square$

Remark 2.3. In electrostatic language: hidden charge is an inverse-source problem for the log-potential measured on the axis, and the perpendicular second moment enters the axial field with a negative sign. The configuration's charge ($2m$ for m hidden pairs) and centre τ are read off from the monopole across several measurement points; m is independently supplied, more cheaply, by Turing-type counting, which the detector complements rather than replaces.

3. THE CERTIFIED RESIDUAL

Fix K and measurement indices $j \leq K$. The detector statistic is

$$(4) \quad R_j = V_j - \sum_{\substack{k \leq K \\ k \neq j}} \frac{2}{\gamma_j - \gamma_k} - \sum_{k \leq K} \frac{2}{\gamma_j + \gamma_k} - T_j,$$

where T_j accounts for the tail $k > K$. Writing $f(u) = 4\gamma_j/(\gamma_j^2 - u^2)$ (so that a zero pair at ordinate u contributes $f(u)$ to the field at γ_j ; f is negative and increasing on (U, ∞) for $U > \gamma_j$), the tail is $\int_U^\infty f \, dN(u)$ with N the zero-counting function and U any point

separating γ_K from γ_{K+1} . Using the exact decomposition $N(u) = \theta(u)/\pi + 1 + S(u)$, where θ is the Riemann–Siegel theta function and $S = \pi^{-1} \arg \zeta$, and integrating the oscillatory part by parts:

$$(5) \quad T_j = \int_U^\infty f(u) \frac{\theta'(u)}{\pi} du - f(U) S(U) - \int_U^\infty f'(u) S(u) du.$$

The three terms are treated as follows.

- (1) The first is a numerical integral of explicit smooth functions.
- (2) The second is *exactly computable*: at a truncation point known (by the verified zero list) to separate γ_K from γ_{K+1} , one has $S(U) = K - \theta(U)/\pi - 1$ exactly. This boundary term is not small — omitting it produces a systematic error of order 10^{-4} at $K = 1000$, comparable to the signals of interest — and its exact evaluation is what makes the band below usable.
- (3) The third is bounded rigorously:

$$|R_j^{\text{true}} - R_j| \leq \varepsilon_j := \int_U^\infty f'(u) B(u) du, \quad B(u) = 0.111 \log u + 0.275 \log \log u + 2.450,$$

using Trudgian’s unconditional bound $|S(u)| \leq B(u)$ for $u \geq e$ [24] and $f' > 0$ on (U, ∞) .

Under completeness of the visible list, the computed R_j therefore lies in the certified band $\pm \varepsilon_j$; any $|R_j| > \varepsilon_j$ certifies a defect — hidden zeros or a missed real zero — and the configuration’s field (Lemma 2.2) is superposed on every R_j . A second integration by parts through explicit bounds on $S_1(u) = \int_0^u S$ would tighten ε_j further; the headroom below has made this unnecessary.

3.1. The pipeline. The full diagnostic, given a (purportedly complete) list $\gamma_1 < \dots < \gamma_K$ and measured V_j at selected j :

- (1) **Existence.** Any $|R_j| > \varepsilon_j$ certifies a defect.
- (2) **Localisation.** The monopole $4m/(x - \tau)$ across several R_j yields the pair count m and centre τ , cross-checked against argument-principle counting.
- (3) **Characterisation.** A nonlinear fit of the exact off-axis model against the exact on-axis model (mirror terms included) selects the better model and returns (τ, σ) or (τ, a) ; the certified version compares fit residuals against ε_j .
- (4) **Margin.** The detected configuration plus the local visible zeros feed the pencil $X + tY$ of Section 5; its real-spectrum threshold is the interaction-corrected local De Bruijn–Newman margin.

4. NUMERICAL RESULTS FOR ζ

All computations below were performed in high-precision arithmetic (`mpmath`, 25–30 significant digits; ordinates stored to 25 digits) and are reproduced by the scripts listed in Section 8.

4.1. Certified null test. With $K = 1000$, $U = 1419.9195$ (the midpoint of $\gamma_{1000}, \gamma_{1001}$) and the exact boundary value $S(U) = +0.153426$:

γ_j	observed R_j	certified ε_j	headroom
103.7255	$-6.3 \cdot 10^{-8}$	$8.0 \cdot 10^{-4}$	$10^{4.1}$
146.0010	$-9.0 \cdot 10^{-8}$	$1.1 \cdot 10^{-3}$	$10^{4.1}$
202.4936	$-1.3 \cdot 10^{-7}$	$1.6 \cdot 10^{-3}$	$10^{4.1}$
321.1601	$-2.2 \cdot 10^{-7}$	$2.6 \cdot 10^{-3}$	$10^{4.1}$
544.3239	$-4.5 \cdot 10^{-7}$	$4.9 \cdot 10^{-3}$	$10^{4.0}$

TABLE 1. Null test on the true Ξ at $K = 1000$. The observed residual is dominated by the 25-digit storage of the ordinate list and the finite precision of the derivative evaluations, four orders below the certified band.

4.2. **Certified thresholds.** At $\gamma \approx 146$ ($\varepsilon = 1.1 \cdot 10^{-3}$), requiring the signal to exceed 2ε :

- *Existence.* A hidden pair’s monopole $4/d$ exceeds 2ε out to $d \approx 1.8 \cdot 10^3$: any hidden zero anywhere near the data window is certifiably visible in a handful of V_j measurements.
- *Sign discrimination.* Off- versus on-axis discrimination needs $4\sigma^2/d^3 > 2\varepsilon$, i.e. $\sigma_{\min} \approx 0.024$ at $d = 1$, 0.067 at $d = 2$, 0.27 at $d = 5$.

For calibration: an isolated off-axis quadruple at height σ forces $\Lambda \gtrsim \sigma^2/2$, so the bound $\Lambda \leq 0.2$ [22, 20] confines any hypothetical off-axis zero to $\sigma \lesssim 0.63$; the interaction corrections computed in Section 5.2 shift this by at most $\sim 15\%$ in the configurations tested. The entire allowed range lies inside the certified sensitive band for $d \leq 5$.

4.3. **Synthetic injections.** To validate steps (2)–(3) of the pipeline in a setting where ground truth is known, we take the first 80 true ordinates, delete zeros #45–46, and insert a hidden configuration at their midpoint $\tau_0 = 134.1271$; the detector sees only the remaining real ordinates and the exact velocities of the modified (finite) multiset. Fitting both exact models (mirror terms included) by least squares:

injected	recovered ($\tau, \cdot$)	rms (correct model)	model separation
off-axis $\sigma = 0.8$	(134.1271235, 0.80000000)	$1.7 \cdot 10^{-11}$	$5.6 \cdot 10^8$
off-axis $\sigma = 0.2$	(134.1271235, 0.20000012)	$9.0 \cdot 10^{-10}$	$6.9 \cdot 10^5$
on-axis $a = 0.3$	(134.1271235, 0.30000003)	$3.0 \cdot 10^{-10}$	$4.8 \cdot 10^6$
on-axis $a = 0.9$	(134.1271235, 0.90000000)	$6.2 \cdot 10^{-14}$	$2.1 \cdot 10^{11}$
empty (control)	—	$\max_j R_j = 3.6 \cdot 10^{-15}$	—

TABLE 2. Injection tests (double precision fits). “Model separation” is the ratio of the wrong model’s best rms to the correct model’s; in every case the correct model and its parameters are recovered, and the empty control is identically zero to machine precision.

Detection *radius* and discrimination *window* differ by design: the certified existence radius ($\sim 10^3$ at $K = 1000$) far exceeds the window where sign discrimination is certified ($d \lesssim 5$ –10); in practice one measures V_j at several zeros bracketing the suspect region, which is exactly the regime where discrimination is strongest.

5. THE DE BRUIJN–NEWMAN FLOW AS A MATRIX PENCIL

For a real polynomial p , define the heat operator on polynomials by $e^{-t\partial^2}p = \sum_{j \geq 0} \frac{(-t)^j}{j!} p^{(2j)}$ (a terminating sum), and

$$\Lambda(p) = \inf\{t \in \mathbb{R} : e^{-s\partial^2}p \text{ is hyperbolic for all } s \geq t\},$$

where *hyperbolic* means all roots real. This is the polynomial De Bruijn–Newman constant: $t > 0$ drives roots onto the real axis, $t < 0$ off it, and the classical Hermite–Poulain–Pólya theory [21, 4] shows hyperbolicity, once acquired, persists (so $\Lambda(p) = \inf\{t : e^{-t\partial^2}p \text{ hyperbolic}\}$).

Theorem 5.1. *Let p be monic of degree n with distinct roots $z_1, \dots, z_n \in \mathbb{C}$, and let (X, Y) be its Calogero–Moser pair,*

$$X = \text{diag}(z_1, \dots, z_n), \quad Y_{jk} = \frac{2}{z_j - z_k} \ (j \neq k), \quad Y_{jj} = \sum_{k \neq j} \frac{2}{z_j - z_k}.$$

Then $[X, Y] + 2I = 2J$ with J the all-ones matrix, so $\text{rank}([X, Y] + 2I) = 1$; and for all $t \in \mathbb{C}$,

$$(6) \quad (e^{-t\partial_z^2}p)(z) = \det(zI - X - tY).$$

Proof. The commutator is immediate: $([X, Y])_{jk} = (z_j - z_k)Y_{jk} = 2$ for $j \neq k$ and 0 on the diagonal, so $[X, Y] = 2(J - I)$.

Both sides of (6) are monic polynomials of degree n in z whose coefficients are polynomials in t , so it suffices to prove equality for t in a real neighbourhood of 0, for generic root data; degenerate configurations follow by continuity.

Left side. Write $H_t = e^{-t\partial_z^2}p$, so $\partial_t H = -\partial_z^2 H$. Implicit differentiation at a simple root $z_j(t)$ gives $\dot{z}_j = H''(z_j)/H'(z_j)$, and for $H = \prod_k (z - z_k)$ one computes $H''(z_j)/H'(z_j) = \sum_{k \neq j} 2/(z_j - z_k)$. Hence the roots obey the first-order system (2), and differentiating (2) along itself — the cross terms cancelling in pairs by antisymmetry — the second-order rational Calogero–Moser system

$$(7) \quad \ddot{z}_j = -8 \sum_{k \neq j} (z_j - z_k)^{-3}.$$

Right side. By the projection method for the rational Calogero–Moser system [1, 10, 14], if $\text{rank}([X, Y] + 2I) = 1$ then the eigenvalues $\lambda_j(t)$ of $X + tY$ solve (7). Their initial values are $\lambda_j(0) = z_j$, and first-order eigenvalue perturbation in the eigenbasis of X gives $\dot{\lambda}_j(0) = Y_{jj}$ — exactly the right side of (2) at $t = 0$. (The pencil trajectory satisfies (2) only at $t = 0$; it is the second-order system (7) that both trajectories share, with matching initial position and velocity.)

Conclusion. Two solutions of the same nonsingular second-order system with identical initial position and velocity coincide until the first collision; on that interval the two monic polynomials have the same roots, hence are equal; and two polynomials in (z, t) agreeing on an open t -interval agree identically. $\square$

Corollary 5.2 (Λ as a spectral threshold). *For real p with distinct roots,*

$$\Lambda(p) = \inf\{t : \text{spec}(X + tY) \subset \mathbb{R}\},$$

and the De Bruijn–Newman flow of p is the straight line $t \mapsto X + tY$ on the Calogero–Moser space $\mathcal{C}_n$.

Proof. By (6), $e^{-t\partial^2}p$ is hyperbolic iff $\text{spec}(X+tY) \subset \mathbb{R}$; apply the persistence of hyperbolicity noted above. $\square$

Corollary 5.3 (quantitative hyperbolicity). *p is hyperbolic iff $X = X+0 \cdot Y$ has real spectrum — trivially — but the pencil upgrades this to a quantitative statement: the local margin of hyperbolicity (the backward collision time of the spectrum) is computable by bisection at one eigendecomposition per step. The positive-definite H with HX Hermitian, whose existence is equivalent to hyperbolicity, is the classical Hermite/Bezoutian form; the pencil does not evade the Hilbert–Pólya heuristic, it furnishes a finite model of it.*

Remark 5.4 (attribution). Identity (6) is Wilson’s tau-function formula: for a Calogero–Moser pair normalised as $\text{rank}([X, Z] - I) = 1$, the tau function of the associated rational KP solution is $\tau(t_1, t_2) = \det(t_1 I + 2t_2 Z - X)$ [26], stated in this determinantal form in [9, §6.3]; with $Z = -Y/2$, $t_1 = z$, $t_2 = t$ this is (6): *the heat flow is the second KP time on polynomial tau functions*. The expectation version over a suitable Gaussian ensemble is finite free probability [12]; (6) is its exact derandomisation, the Calogero–Moser matrix Y being the deterministic representative that the expectation happens to equal (cf. [8] for the random-matrix heat-flow circle of ideas). The root dynamics (2) in the De Bruijn–Newman context is [23, Theorem 11] and, as pole dynamics of rational KP/Burgers solutions, goes back to [1]. What we claim as new is only the De Bruijn–Newman reading: Corollaries 5.2 and 5.3, and their use as a margin meter below.

Remark 5.5 (a worked benchmark). $e^{-t\partial^2}z^d$ is (a rescaled) Hermite polynomial, hyperbolic exactly for $t \geq 0$, so $\Lambda(z^d) = 0$; since the probabilists’ Hermite polynomial is $\text{He}_d = e^{-\frac{1}{2}\partial^2}z^d$, the semigroup law gives $\Lambda(\text{He}_d) = -\frac{1}{2}$ for every d . Λ scales quadratically under affine root maps, and the root variance of He_d is $d - 1$, so at unit root variance $\Lambda = -1/(2(d - 1))$ — reproduced by the pencil to bisection tolerance (Section 5.1). Similarly $\Lambda(z^2 + y_0^2) = y_0^2/2$ exactly, the pencil recovering it to $2 \cdot 10^{-13}$.

5.1. Numerical verification of the pencil. Independently of the proof: the identity (6) was checked by coefficient comparison for random real and conjugate-pair root data at $n = 2, \dots, 7$ and $t \in \{0.13, -0.4, 0.7 + 0.2i\}$, with maximal relative coefficient error $5 \cdot 10^{-12}$ (double precision, conditioning-limited) and rank-one residual $\leq 2 \cdot 10^{-15}$; $\Lambda(z^2 + y_0^2) = y_0^2/2$ and the Hermite benchmark of Remark 5.5 are reproduced; and on five two-pair configurations $\pm(\tau_1 \pm i\sigma_1), \pm(\tau_2 \pm i\sigma_2)$ the pencil threshold agrees with direct root tracking of the heat flow to $2 \cdot 10^{-11}$ (e.g. $(\tau, \sigma) = (0, 1.0), (3.0, 0.5)$ gives $\Lambda = 0.42976$ against the isolated-pair value 0.5 — interactions matter, which is the point of the margin meter).

5.2. The margin meter. Step (4) of the pipeline: the detected configuration plus its local environment feeds the pencil, whose threshold is the interaction-corrected local De Bruijn–Newman margin — the off-axis dual of the Lehmer-pair lower bounds on Λ of Csordas–Smith–Varga [5] (close *real* pairs force Λ up; off-axis pairs force it up through their height). For a quadruple $\pm(\tau_0 \pm i\sigma)$ at $\tau_0 = 134.127$ with the 6 nearest visible zeros on each side (and mirrors) included:

6. PORTABILITY: THE DIRICHLET BETA FUNCTION

Nothing above used more than the Hadamard structure (1), evenness, and reality, so the method applies to $\Xi_L(z) = \Lambda_L(\frac{1}{2} + iz)$ for any self-dual Selberg-class L -function. As a demonstration take $L(s, \chi_4)$ (the Dirichlet beta function), with completed form $\Lambda_\beta(s) = (\pi/4)^{-(s+1)/2} \Gamma(\frac{s+1}{2}) L(s, \chi_4)$: Ξ_β is even, entire and real, with first zeros at 6.0209, 10.2438, 12.9881, $\dots$

σ	isolated quadruple	with 2×6 local zeros	$\sigma^2/2$
0.8	0.319994	0.273756	0.320000
0.4	0.080000	0.076504	0.080000
0.2	0.020000	0.019769	0.020000

TABLE 3. Pencil thresholds (local Λ margins). The isolated quadruple reproduces $\sigma^2/2$; the local Coulomb environment lowers the margin by up to $\sim 15\%$ at $\sigma = 0.8$, an interaction correction invisible to the isolated-configuration heuristic.

(recomputed). With 70 zeros found by sign scanning and the smooth density $dN = (2\pi)^{-1} \log(2u/\pi) du$ (verified empirically: $N/T = 0.4533$ measured against 0.4561 smooth at $T = 75$), the null test gives residuals

$$+4.4 \cdot 10^{-4}, \quad +7.9 \cdot 10^{-4}, \quad +1.2 \cdot 10^{-3}$$

at $\gamma_{11}, \gamma_{21}, \gamma_{31}$ — the same quality as the zeta prototype at comparable list length with an (uncertified) smooth-density tail. Full certification for Dirichlet L -functions requires only the χ -analogue of the exact $\theta/\pi + 1 + S$ decomposition and an explicit bound on $|S(u, \chi)|$, both available in the literature; nothing else changes. The practical interest is precisely here: verified heights for L -functions are far below those for ζ , and the detector’s inputs — two derivative evaluations of the completed function per measured zero — are available wherever zeros are computed.

7. LIMITATIONS

- (1) The certificates are rigorous modulo floating point: V_j , the theta integrals and ε_j are computed in high-precision arithmetic, not interval arithmetic. An interval port (e.g. ARB) is mechanical and is the natural next engineering step.
- (2) The method certifies and diagnoses; it does not prove. If RH is true the detector will only ever output null results.
- (3) The novelty claims are confined to: the certified residual decomposition (5) with its exactly computable boundary term; the sign discriminant of Lemma 2.2 as a practical classifier; and the De Bruijn–Newman reading of Wilson’s formula (Corollaries 5.2–5.3). The identity (6) itself is Wilson’s [26]; the velocity V_j is classical as a flow velocity [23]. We have not found Corollary 5.2 stated in the De Bruijn–Newman literature (in particular in the Csordas school), but a definitive novelty claim would require a more systematic search than we have performed.

8. REPRODUCIBILITY

All numerics are reproduced by short Python scripts (`numpy`, `scipy`, `mpmath`) accompanying this paper as ancillary files: `h1_zeros1000.py` (ordinate cache), `h3_certified.py` (Table 1 and the thresholds of Section 4.2), `d1_synthetic.py` (Table 2), `v1_pencil.py` (Section 5.1), `d4_margin_meter.py` (Table 3), `h4_beta.py` (Section 6), together with the zero caches `zeros1001.json` and `beta_zeros.json`.

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REFERENCES

- [1] H. Airault, H. P. McKean, J. Moser, *Rational and elliptic solutions of the Korteweg–de Vries equation and a related many-body problem*, Comm. Pure Appl. Math. **30** (1977), 95–148.
- [2] M. F. Atiyah, V. G. Drinfeld, N. J. Hitchin, Yu. I. Manin, *Construction of instantons*, Phys. Lett. A **65** (1978), 185–187.
- [3] A. R. Booker, *Turing and the Riemann hypothesis*, Notices Amer. Math. Soc. **53** (2006), 1208–1211.
- [4] N. G. de Bruijn, *The roots of trigonometric integrals*, Duke Math. J. **17** (1950), 197–226.
- [5] G. Csordas, W. Smith, R. S. Varga, *Lehmer pairs of zeros, the de Bruijn–Newman constant Λ , and the Riemann hypothesis*, Constr. Approx. **10** (1994), 107–129.
- [6] N. Dorey, T. J. Hollowood, V. V. Khoze, M. P. Mattis, *The calculus of many instantons*, Phys. Rep. **371** (2002), 231–459; [hep-th/0206063](#).
- [7] M. Griffin, K. Ono, L. Rolin, D. Zagier, *Jensen polynomials for the Riemann zeta function and other sequences*, Proc. Natl. Acad. Sci. USA **116** (2019), 11103–11110.
- [8] B. C. Hall, C.-W. Ho, *The heat flow conjecture for polynomials and random matrices*, Lett. Math. Phys. **115** (2025), art. 60; [arXiv:2202.09660](#).
- [9] A. Kasman, R. Milson, M. Gekhtman, *Orthogonality with respect to the Hermite product, KP wave functions, and the bispectral involution*, SIGMA **22** (2026), 036, 23 pp.; [arXiv:2511.11813](#).
- [10] D. Kazhdan, B. Kostant, S. Sternberg, *Hamiltonian group actions and dynamical systems of Calogero type*, Comm. Pure Appl. Math. **31** (1978), 481–507.
- [11] D. H. Lehmer, *On the roots of the Riemann zeta-function*, Acta Math. **95** (1956), 291–298.
- [12] A. W. Marcus, D. A. Spielman, N. Srivastava, *Finite free convolutions of polynomials*, Probab. Theory Related Fields **182** (2022), 807–848; [arXiv:1504.00350](#).
- [13] H. L. Montgomery, *The pair correlation of zeros of the zeta function*, in: Analytic Number Theory, Proc. Sympos. Pure Math. **24**, Amer. Math. Soc., 1973, 181–193.
- [14] J. Moser, *Three integrable Hamiltonian systems connected with isospectral deformations*, Adv. Math. **16** (1975), 197–220.
- [15] H. Nakajima, *Lectures on Hilbert Schemes of Points on Surfaces*, Univ. Lecture Ser. **18**, Amer. Math. Soc., 1999.
- [16] N. Nekrasov, A. Schwarz, *Instantons on noncommutative $\mathbb{R}^4$, and $(2, 0)$ superconformal six dimensional theory*, Comm. Math. Phys. **198** (1998), 689–703.
- [17] C. M. Newman, *Fourier transforms with only real zeros*, Proc. Amer. Math. Soc. **61** (1976), 245–251.
- [18] A. M. Odlyzko, *On the distribution of spacings between zeros of the zeta function*, Math. Comp. **48** (1987), 273–308.
- [19] D. J. Platt, *Isolating some non-trivial zeros of zeta*, Math. Comp. **86** (2017), 2449–2467.
- [20] D. J. Platt, T. S. Trudgian, *The Riemann hypothesis is true up to $3 \cdot 10^{12}$* , Bull. Lond. Math. Soc. **53** (2021), 792–797; [arXiv:2004.09765](#).
- [21] G. Pólya, *Über die algebraisch-funktionentheoretischen Untersuchungen von J. L. W. V. Jensen*, Kgl. Danske Vid. Sel. Math.-Fys. Medd. **7** (1927), 3–33.
- [22] D. H. J. Polymath, *Effective approximation of heat flow evolution of the Riemann ξ function, and a new upper bound for the de Bruijn–Newman constant*, Res. Math. Sci. **6** (2019), art. 31.
- [23] B. Rodgers, T. Tao, *The de Bruijn–Newman constant is non-negative*, Forum Math. Pi **8** (2020), e6, 62 pp.
- [24] T. S. Trudgian, *An improved upper bound for the argument of the Riemann zeta-function on the critical line II*, J. Number Theory **134** (2014), 280–292; [arXiv:1208.5846](#).
- [25] A. M. Turing, *Some calculations of the Riemann zeta-function*, Proc. Lond. Math. Soc. (3) **3** (1953), 99–117.
- [26] G. Wilson, *Collisions of Calogero–Moser particles and an adelic Grassmannian*, Invent. Math. **133** (1998), 1–41.